Technical Reference Manual

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Independent Engineering Design and Process Modeling Project
August 2026 (v1.0)
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Bioethanol Production, Biochemical Pathways, Thermodynamic Modeling, Kinetic Modeling, Python Simulation, Design of Experiments

Introduction

This manual provides a comprehensive understanding surrounding the fundamental and practical aspects of ethanol production using *Saccharomyces cerevisiae (*var. *Voss Kveik)* which can be used in various industries. The manual serves as an engineering reference for designing and optimizing biochemical processes. It provides the grounding which connects practical work to theoretical and can be referenced in future investigations focused on different aspects of fermentation.

Biochemical Fundamentals

Carbohydrates

The primary source of fermentable sugars are found in malted barley which has an average chemical composition of 57.7 - 60.7% starch on a dry matter basis (Jasmina Lukinac & Marko Jukić, 2022, sec. 4). (Oscarsson et al., 1996, p. 163). Starch is a polysaccharide made up of many monosaccharides bonded together. The simplest carbohydrates are monosaccharides which vary widely such as the amount of carbon atoms that make them up, generally being named by a prefix that indicates the number of carbons and a suffix -ose indicating it is a saccharide. The hexose, D-glucose is the most abundant saccharide found in nature and is found in cyclic ring structures. Carbohydrates serve a few purposes for microbes such as structural as with pectin, cellulose or hemicellulose, they also serve as an energy source for organisms in the form of storage molecules like glycogen or starch or free monosaccharides like maltose, glucose or maltotriose. They are also found in the genetic material of organisms, in the form of ribose and deoxyribose monosaccharides, allowing for information to propagate throughout generations.

Enzymes

Enzymes are biological molecules which increase the rate of reactions for organisms without being consumed in the reaction, acting as a biological catalyst allowing for essential metabolic reactions to take place. There are six primary categories which enzyme functions can be organized into; hydrolases, oxidoreductases, transferases, lyases, isomerases and ligases (Robinson, 2015).

The production of bioethanol involves a few key enzymes; α-amylase, β-amylase, limit dextrinase, pyruvate decarboxylase and alcohol dehydrogenase. During wort preparation, α-amylase, β-amylase and limit dextrinase are key enzymes present in malted barley (Biolaxi Enzyme Team, 2024) and work to hydrolyse different bonds in starch producing fermentable sugars like maltose or glucose. These sugars are then consumed by *S. cerevisiae* which employs metabolic enzymes, pyruvate decarboxylase and alcohol dehydrogenase in order to produce ethanol (van Aalst et al., 2022).

Enzyme reaction mechanisms can be understood through a few key concepts; the active site is a specific space on the enzyme which is suitable for substrate binding and conversion into products, substrates that bind with the enzyme form complexes where the interaction has a more stable transition state than the uncatalyzed conversion and the active site can conform to the substrate shape avoiding the need to have very specific orientation or geometry for a reaction to proceed (Tang, 2024).

Enzyme performance largely depends on various outside factors; the concentration of substrate influences the rate of enzyme-substrate complex formation because at low concentrations, less substrates will find their way to an active site, while at high concentrations, all available active sites may be taken up causing a plateau in reaction rate. Temperature and pH affect the structure of the enzyme active site, outside of the optimal temperature or pH, the enzyme denatures as bonds are broken. At low temperatures, substrates and enzymes lose kinetic energy and the frequency of their interactions decreases. The presence and concentration of cofactors or inhibitor molecules can also affect the enzyme's performance (Tang, 2024).

Glycolysis

One of the oldest metabolic processes evolved and preserved in organisms is the breakdown of a glucose molecule into two pyruvate molecules, producing two ATP molecules and two NADH molecules, which is called glycolysis. A few other monosaccharides can be used instead of glucose such as fructose and galactose which are also ingested by the organism (Chandel, 2021) .

$$Glucose \rightarrow 2Pyruvate + 2ATP + 2NADH$$

The presence or absence of oxygen determines what happens after glycolysis to the pyruvate molecules within *S. cerevisiae*. Under anaerobic conditions, the enzymes pyruvate decarboxylase and alcohol dehydrogenase convert pyruvate into ethanol and carbon dioxide in order to oxidize two molecules of NADH from glycolysis into NAD^+ to continue the glycolysis cycle (Ishtar Snoek & Yde Steensma, 2007).

Ethanol Synthesis

Ethanol synthesis in *S. cerevisiae* is an anaerobic metabolic pathway which is followed when there is an absence of molecular oxygen. It consists of glycolysis then alcoholic fermentation. First glycolysis turns glucose into 2 pyruvate molecules, 2 ATP molecules and 2 NADH molecules (Chandel, 2021). Pyruvate undergoes decarboxylation which is catalyzed by pyruvate decarboxylase producing acetaldehyde and CO_2. However NADH still needs to be oxidized into NAD^+ in order for glycolysis to continue producing energy essential for survival, this is done by the reduction of acetaldehyde into ethanol (Rymuszka & Gorczynska, 2026; van Aalst et al., 2022).

Secondary Metabolism

An important feature of alcoholic fermentation with *S. cerevisiae* is the influence of secondary metabolism, this type of metabolism is not essential for growth or reproduction but it begins once the organism has completed its logarithmic growth phase. Secondary metabolites have a wide range of chemical structures and biological activities and their purpose is often to improve the survival of the organism by providing advantages within its ecosystem such as against competing bacteria or repelling predator organisms etc. They are derived from unique pathways using primary intermediates and metabolites and different enzymes. The type of secondary metabolites and quantity are controlled by the genes of the organism and is strain specific (Vining, 1992).

General strains of *S. cerevisiae* produce a range of secondary metabolites; higher (fusel) alcohols (Loviso & Libkind, 2019), esters (Saerens et al., 2010), vicinal diketones (Stewart, 2017), carbonyl compounds, sulfur compounds, organic acids, phenolic compounds and fatty acids. Each of which affect the separation of ethanol from the fermented mixture or affect the sensory profile in alcoholic beverages.

Fusel alcohols are alcohols with more than 2 carbon atoms produced by yeast during alcoholic fermentation, a few major compounds are: isobutanol, 3-methylbutanol and propanol (Loviso & Libkind, 2019). Fusel alcohols are formed via the Ehrlich pathway which was observed to arise from the breakdown of amino acids where increasing concentration of certain amino acids correlated to an increase in a certain fusel alcohol (Hazelwood et al., 2008). The Ehrlich pathway can be summarized briefly by the following steps;

Transamination - A free amino acid enters the cytoplasm where enzymes (BAT2, ARO8, ARO9) convert it into α-keto acids, freeing up ammonia for yeast cells to use for protein synthesis (Hazelwood et al., 2008).

Decarboxylation - CO_2 is removed from the α-keto acids by decarboxylase enzymes (ARO10, PDC1, PDC5, PDC6), forming 'fusel aldehydes' (Hazelwood et al., 2008).

Oxidation - The 'fusel aldehydes' are then oxidized by alcohol dehydrogenase (ADH), and other redundant enzymes, into 'fusel acids', using NAD^+ and releasing NADH and H^+ (Hazelwood et al., 2008).

Reduction - Reduction by alcohol dehydrogenase turns 'Fusel aldehydes' into 'fusel alcohols' using NADH, H^+ and restoring NAD^+ (Hazelwood et al., 2008).

A brief overview of this process is shown in *Figure 1* below using the amino acid, valine, as an example.

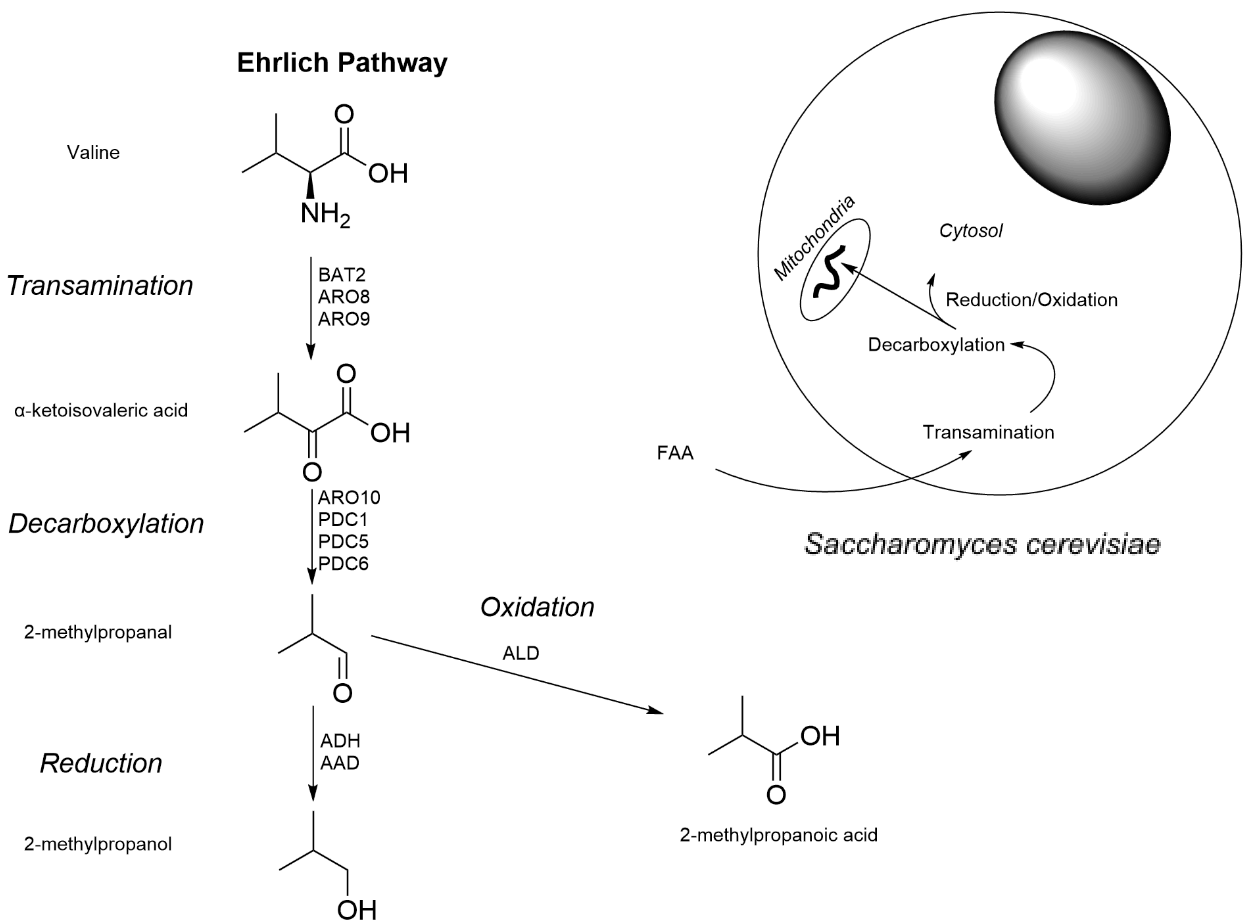

Figure 1 - Ehrlich pathway in S. cerevisiae using valine as an example

Esters are another secondary metabolite formed by *S. cerevisiae*, their production affects downstream separation processes and value added chemical products when producing bioethanol, additionally esters have a major impact on the organoleptic profile of alcoholic beverages. Typically, the esters that are produced are categorized into two groups with their distinctive major components (Stewart, 2017), these two groups are acetate esters where the alcohol group is ethanol or other higher alcohols (e.g. ethyl acetate, isoamyl acetate, isobutyl acetate) and medium-chain fatty acid ethyl esters (MCFA ethyl esters) where the alcohol group is ethanol and the acid group is a fatty acid (e.g. ethyl hexanoate, ethyl octanoate, ethyl decanoate) (Saerens et al., 2010).

Ester biosynthesis is a byproduct of three metabolic processes, alcoholic fermentation, fatty acid synthesis and the Ehrlich pathway. In the alcoholic fermentation pathway, acetaldehyde molecules can be converted into acetate by aldehyde dehydrogenase then into acetyl-CoA by acetyl-CoA synthase. Some acetyl-CoA can react with fusel alcohols from the ehrlich pathway, catalyzed by alcohol acyltransferase (ATF1/ATF2), resulting in acetate ester formation. Acetyl-CoA in the cytosol is converted by Acetyl-CoA carboxylase (ACC1) and CO_2 into malonyl-CoA and a condensation/reduction cycle is started causing 2 carbons to be added to the chain each cycle Palmitoyl-CoA when the cycle ends. Some MCFAs (6-10 carbons) are not used by fatty acid synthase complex (FAS1/FAS2) and instead are released prematurely then later react with ethanol catalyzed by ethanol hexanoyl transferase and acyl-CoA: ethanol O-acyltransferase (EHT1/EEB1) forming ethyl esters (Saerens et al., 2010). *Figure 2* below shows an overview of the biosynthetic process.

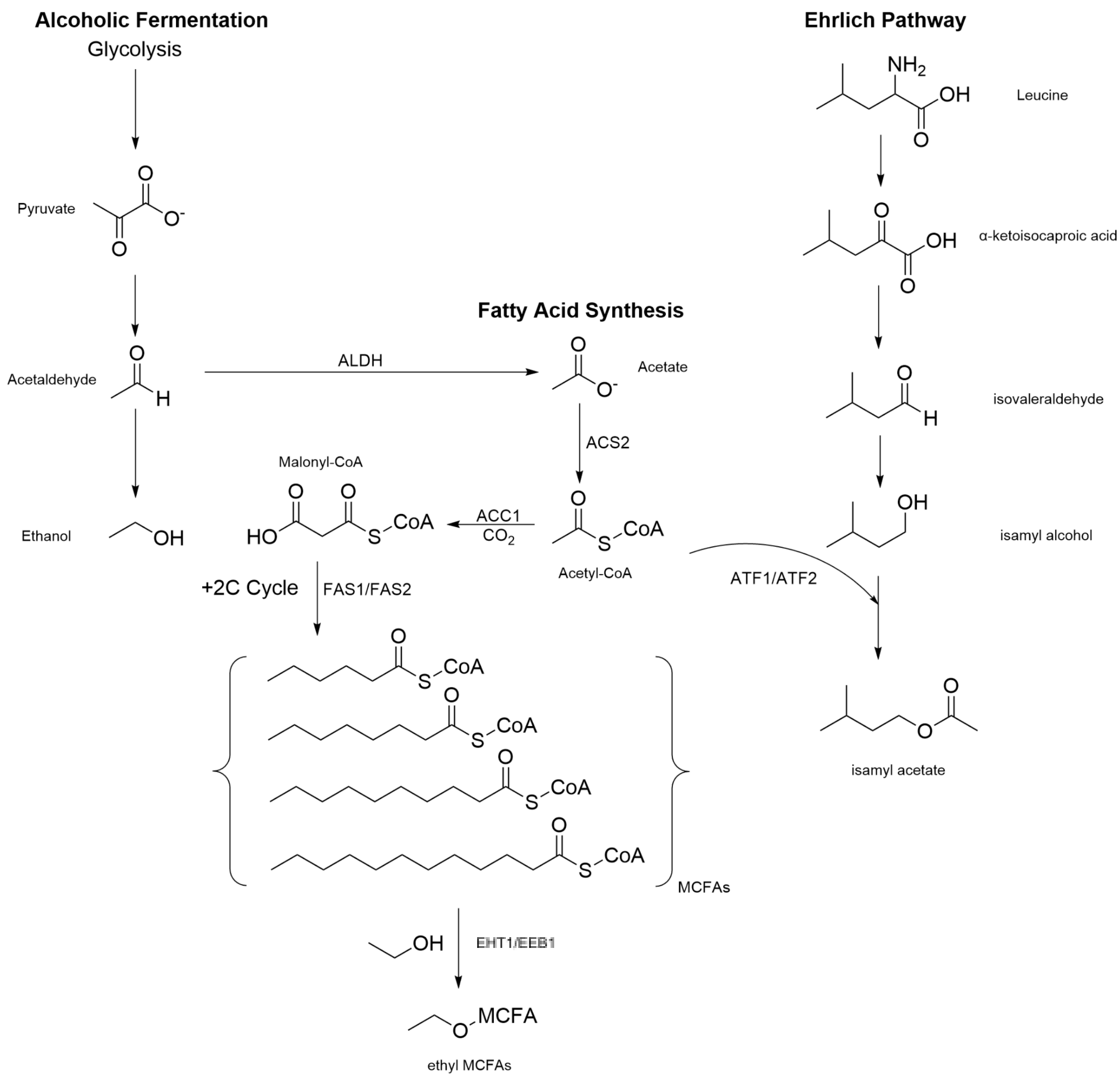

Figure 2 - Example of a biosynthetic pathway resulting in ester formation for S. cerevisiae

From the alcoholic beverage industry, the sensory profile is incredibly important for quality assurance for businesses, *Table 1* is shown below linking compounds mentioned previously as examples to commonly perceived aroma.

Table 1

Commonly perceived aromas in a few major secondary metabolites

Compound	Aroma
isobutanol	Solvent, alcoholic, slightly sweet, pungent
3-methylbutanol	Pungent, alcoholic, solvent, whisky-like, slightly banana

Propanol	Solvent, alcoholic, musty, faintly fruity
ethyl acetate	Light fruity, pear, sweet, solvent
isoamyl acetate	Banana, pear, citrus, sweet
isobutyl acetate	Fruity, sweet apple, raspberry
ethyl hexanoate	Pineapple, green apple, anise
ethyl octanoate	Floral, ripe apple, brandy, oily
ethyl decanoate	Grape, floral, waxy

Kinetic Models

Overview

As with any chemical reaction, the kinetics play a large role in engineering a process for optimal performance. For the stout brewing process, we need a practical way to understand the fermentation process where sugars are consumed anaerobically by *S. cerevisiae*, producing ATP along with ethanol and carbon dioxide. According to D'Amore et al. (1989), *S. cerevisiae* was found to have a much higher affinity for glucose rather than fructose both of which are found in the disaccharide sucrose, thus for the purpose of simplicity, only the metabolism of glucose is considered, however it is important to note that many other fermentable sugars may be present in the wort, such as; maltose, maltotriose, sucrose and possibly fructose each of which have different uptake rates. Additionally, non-fermentable sugars are present; dextrins and pentoses among others, which impact the specific gravity measurements by increasing the measured concentration of sugar in the mixture (Beer-o-Meter, 2025). The simplified fermentation reaction is shown below:

$$C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2 + ATP$$

Kinetic Models

Various kinetic models exist to investigate different parameters, such as, microbial population growth rate, maximum population size, substrate consumption kinetics and inhibitor effects; however, each model has limitations tied to the assumptions made or variables ignored so it is important to use them in combination with each other to address limitations of each. Two kinetic models were chosen in order to compare the modeling of substrate consumption between an empirical and a mechanistic model allowing for a detailed understanding of the fermentation process.

It is important to understand that there are three distinct classes of mathematical models; empirical models, semi-empirical models and mechanistic models, each of which has important advantages and disadvantages. Empirical models serve as a great way to describe the relationship between experimental data however it does not explain why certain results are obtained so extrapolation of results is uncertain. Mechanistic models offer insight into the dynamics of a system allowing more detailed understanding of the causes of a systems behavior, these models are usually open to be extended with other important factors to the process because they rely on certain principles such as mass balances or energy balances, however these are the most computationally expensive models and are more geared towards research rather than applications. Semi-empirical models fall between the two, offering some process understanding but having fitted parameters estimated from experiments (García-Rodríguez et al., 2022, pp. 5-6).

Empirical Model: Modified Gompertz

The Gompertz model is an empirical model that describes the way a population grows, taking into account age based mortality, it takes the form of $L_x = d \cdot g^{q^x}$ (Gompertz, 1815, p. 518) however this original form was modified by Zwietering et al in the late 1900s and the original form was adapted to microbial populations, revealing lag time, growth rate and maximum population (Zwietering et al., 1990). For fermentation, the key kinetic parameters were adapted to reveal information on the sugar consumption, below is the adapted model where S(t) is the sugar consumed as a function of time in (g L^{-1}), R_{max} is the maximum sugar consumption rate in (g L^{-1} h^{-1}), S_{max} is the maximum sugar that can be consumed in (g L^{-1}), λ is the lag phase duration in (h), t is the time in (h) and e is Euler's number. The modified Gompertz is particularly useful because with few parameters it is possible to capture the sigmoidal nature of fermentation fitted through experimental data.

Modified Gompertz Model $$S(t) = S_{max} \cdot exp[- exp(\frac{R_{max}e}{S_{max}}(\lambda - t) + 1)]$$

Due to the nature of the model being empirical it is possible to fit it onto experimental results by substituting the calculated sugar consumed from specific gravity (SG) data and setting the S_{max} equal to that value, then the lag time is determined roughly by the appearance of CO_2 bubbles from the start of fermentation, then from data measurements determining the R_{max} is possible by using the second derivative of the model to find the inflection point and then using the equation that Zwietering et al. (1990) derived in order to find the R_{max} which aligns with experimental observations;

$$t_i = \lambda + \frac{S_{max}}{R_{max}e} \Rightarrow R_{max} = \frac{S_{max}}{e(t_i - \lambda)}$$

Semi-Empirical Model: Modified Monod-Logistic Model

The second kinetic model used was a modified monod-logistic model which predicts sugar consumption over time taking into account various biological processes such as biomass growth, carrying capacity and product inhibition. The benefit of using this model is that it factors in real limiting factors and driving forces in fermentation giving rise to accurate predictions.

Modified Monod-Logistic Model $$\frac{dS}{dt} = - R \cdot X$$

$$\frac{dE}{dt} = Y_{E/S}(-\frac{dS}{dt})$$

$$\frac{dX}{dt} = \mu X(1 - \frac{X}{X_{max}})$$

Where

$$R = R_{max}\frac{S}{K_s+S}$$

And

$$\mu = \mu_{max}\frac{S}{K_s+S}(1 - \frac{E}{E_{max}})^{\beta}$$

This model uses coupled differential equations in order to model the fermentation process. R (g L^{-1} h^{-1}) is the instantaneous rate of consumption of sugar while R_{max} (g L^{-1} h^{-1}) is the maximum rate of consumption observed experimentally. The instantaneous growth rate of yeast is represented by μ (g L^{-1} h^{-1}) and the maximal growth rate is given by μ_{max} (g L^{-1} h^{-1}). The growth rate and rate of consumption depend on the concentration of sugar represented by S (g L^{-1}) and the half velocity constant K_s (g L^{-1}). The growth rate of yeast is limited by the concentration of ethanol which causes protein misfolding and increasing membrane fluidity (Kitagaki, 2008), this is modelled by the logistic term, $(1 - \frac{E}{E_{max}})^{\beta}$, where E (g L^{-1}) is representative of ethanol concentration, and E_{max} (g L^{-1}) is the maximum tolerable concentration (Aiba et al., 1968). β is an empirical inhibition exponent dictating the strength of the inhibition. The growth of yeast is measured by the biomass term X (g L^{-1}) and is also inhibited by a biomass carrying capacity term which is specific to the strain of yeast and the resources available in the environment, this is represented by X_{max} (g L^{-1}). $Y_{E/S}$ is the ethanol yield coefficient which is calculated by $\frac{\Delta E}{\Delta S}$ and gives the portion of sugar turned into ethanol.

Despite being mechanistic and taking into account biological factors propagating fermentation, the model still has limitations to its accuracy across various initial conditions. The biomass growth of yeast requires nutrients like free amino nitrogen groups (FANs) in order to synthesize essential proteins, so over the course of the fermentation the concentration of FANs decreases, limiting the rate of fermentation. Another limitation is the assumption that the system is fully anaerobic from the beginning; however this is not the case in reality because oxygen is present at the start before being consumed and anaerobic conditions are reached. Additionally, values from literature are taken for some kinetic constants which may reduce the accuracy to the specific strain of yeast used in the fermentation and these kinetic constants used across both chosen models are assumed to be constant however in reality they vary across temperature.

A few parameters required to model substrate consumption during fermentation are difficult to measure using basic homebrew equipment, therefore literature values for *S. cerevisiae* can be taken for modeling purposes. This particular fermentation process employs the use of LalBrew Voss™ Kveik which is a Norwegian farmhouse strain of yeast known for its abnormally high fermentation rates and very high optimal temperature of 35-40℃ with a remarkable lack of off-flavors. Additionally it has a high alcohol tolerance of 12% ABV and an attenuation of 76-82 % (Lallemand Inc., 2021) . Due to the difficulty of finding literature sources for strain-specific kinetic parameters of *S. cerevisiae var. Voss Kveik*, baseline values were adopted to establish the models with reasonable accuracy, these are shown below in *Table 2.*

Table 2

Kinetic Parameters Chosen for Modeling Voss Kveik

Parameter	Symbol	Value	Source
Maximum Growth Rate	μ_{max}	0.45 h^{-1}	(Bioprocess Tools, 2026)
Half Saturation Constant	K_s	0.025 g L^{-1}	(Bioprocess Tools, 2026)
Carrying Capacity	X_{max}	8 g L^{-1}	-
Inhibition Exponent	β	2	Fitted Parameter

R_{max} Derivation

Other kinetic parameters which are used in the Gompertz model can be worked out by processing the raw data from each batch. For batches where airlock bubble rate is measured, molar flow rate of CO_2 can be worked out by first taking note of the airlock dimensions to find the bubble volume, V_{bbl}, then calculating the pressure difference, ΔP, required to produce one bubble, then calculating the molar volume, $V_{n,\,CO2}$, inside the fermenter. With those values it is possible to find the moles per bubble, n_{bbl}, which is used alongside airlock bubble rate data to calculate the rate of sugar consumption and R_{max}. This data can also be used to find the lag time, λ, and using OG and FG the maximum sugar consumption parameter can be found. The derivation is shown below.

$$\Delta P = \rho g h$$

Where ρ is the density of liquid in the airlock, g is the gravitational acceleration, h is the height.

$$V_{bbl} = \frac{1}{8}\pi^2 D^2 (R - r)$$

Where D is the diameter of the airlock at the elbow bend, R is the radius from the center of the bend to the furthest part of the elbow and r is the radius to the nearest part of the elbow.

$$\frac{PV}{n} = RT \Rightarrow V_{n,\,bbl} = \frac{P_{atm} V_{n,\,CO2}}{P_{bbl}}$$

The ideal gas law is rearranged and the atmospheric conditions are equated to the fermenter conditions where V_n represents molar volume.

$$n_{bbl} = \frac{V_{bbl}}{V_{n,\,bbl}}$$

$$\varphi_{n,\,CO2} = Q n_{bbl}$$

Where Q is the bubble rate in h^{-1}.

$$\frac{dn_{glucose}}{dt} = -\frac{1}{2}\varphi_{n,\,CO2} \Rightarrow \frac{d\left(\frac{C_{m,glu} V}{M_{glu}}\right)}{dt} \Rightarrow \frac{dC_{m,\,glu}}{dt} = -\frac{M_{glu}}{2V}\varphi_{n,\,CO2}$$

The rate of CO_2 production is proportional to the rate of glucose consumption in moles so the values are converted to glucose consumption in (g L^{-1} h^{-1}). Converting all the bubble rate data into glucose consumption and finding the maximum value allows an estimate of the R_{max} kinetic parameter which can be compared to the calculated value from the formula that Zwietering et al. (1990) derived.

Temperature Control

Overview

During the beginning of the brewing process, the grains must be steeped in hot water allowing solutes to dissolve forming a wort. Starch is the target component for this stage because it will be broken down by α-amylase and β-amylase into fermentable sugars and turned into alcohol by yeast. α-amylase and β-amylase are naturally present in malted barley, however, optimal conditions for the enzymes are between 70-75℃ for α-amylase (Busch, 2025) and 55-60℃ for β-amylase with operation up to 70℃ (Creative Enzymes, 2026; Silano et al., 2017). The chosen temperature range for this steeping procedure is 62-72℃ due to limitations in accuracy with the use of an induction stove.

To maintain this temperature range, the induction stove will be turned on until the temperature of the water reaches 72℃. Then the stove is turned off and only turned back on when the temperature of the wort approaches 62℃. For this, a time estimate for the temperature change is required to ensure optimal temperature control.

Modeling

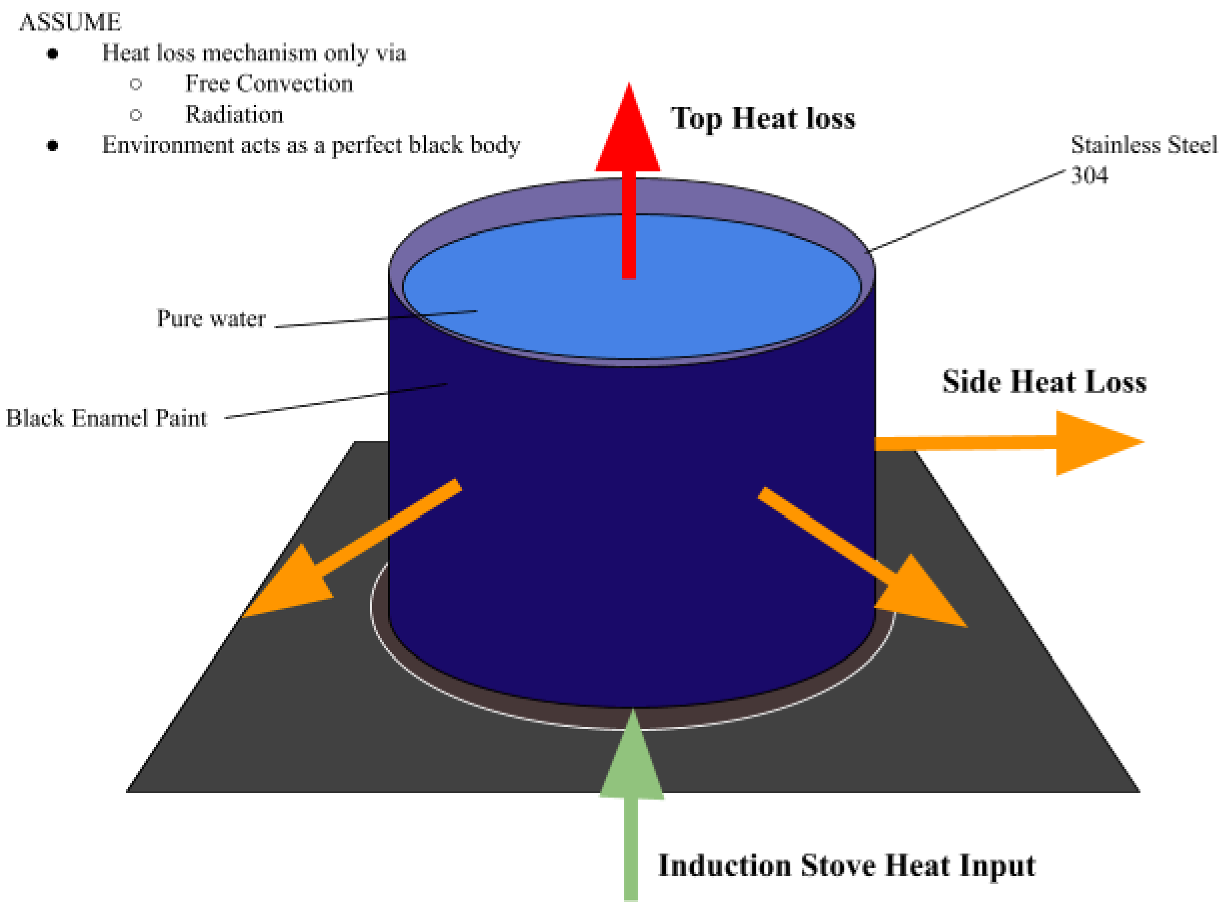

Figure 3 - System used to model heat losses

Figure 3 depicts the model that was used to help estimate the time it takes for the wort to decrease in temperature from 72℃ to 62℃. The governing equations being used are as follows;

$$V\frac{dE}{dt} = \Phi_{H,in} - \Phi_{H,out} + qV \Rightarrow \frac{dE}{dt} = \Phi_{H,in} - \Phi_{H,out}$$

Where the stove heat inflow is given by;

$$\Phi_{H,in} = \frac{m_{total} C_{p\,total} \Delta T}{t}$$

And the heat losses are;

$$\Phi_{H,out} = \Phi_{FC} + \Phi_{Rad}$$

$$\Phi_{H,out} = \Phi_{FC,\,top} + \Phi_{FC,\,side} + \Phi_{Rad,\,top} + \Phi_{Rad,\,side}$$

Free convection heat loss and radiation heat loss can be estimated using the equations;

$$\Phi_{FC} = h(T_{avg} - T_{env})(A_{side} + A_{top})$$

$$\Phi_{Rad} = \sigma(T_{avg}^4 - T_{env}^4)(e_{side}A_{side} + e_{top}A_{top})$$

Where the specifications of the pot were found to be:

Diameter $D = 24.2\ cm$ Side Area $A_{side} = \pi D h = 0.0806\ m^2$

Height $h = 10.6\,cm$ Top Area $A_{top} = \frac{1}{4}\pi D^2 = 0.046\,m^2$

Wall Thickness $\delta = 0.3\,cm$ Pot mass $m_{pot} = \rho\pi\delta(h + \frac{1}{2}D - \frac{1}{4}\delta) = 0.1706\,kg$

Table 3

Thermodynamic Modeling Parameters Used

Constant	Value	Source
	Stainless Steel 304	
λ_{SS304} (W m^{-1} k^{-1})	16.2	(304 Stainless Steel, n.d.)
$C_{p\ SS304}$ (J kg^{-1} k^{-1})	500	(304 Stainless Steel, n.d.)
ρ_{SS304} (kg m^{-3})	7930	(304 Stainless Steel, n.d.)
e_{SS304}	0.80	(The Engineering ToolBox, 2003, Black Enamel Paint)
	Water	
λ_{H2O} (W m^{-1} k^{-1})	0.6	(Cengel & Ghajar, 2015)
$C_{p\ H2O}$ (J kg^{-1} k^{-1})	4184	(Cengel & Ghajar, 2015)
ρ_{H2O} (kg m^{-3})	1000	(Cengel & Ghajar, 2015)
e_{H2O}	0.963	(Cengel & Ghajar, 2015)
	Universal	
h (W m^{-2} k^{-1})	6.63 *	Experimental
σ (W m^{-2} k^{-4})	5.67e-08	(Santos, n.d.)
T_{env} (℃)	23	Assumption

* *Convective heat transfer coefficient, h, for still air varies between 2-25 W m^{-2} k^{-1}* (Incropera et al., 2011) *which greatly affects the calculated rate of heat loss therefore a representative value was chosen using the average of a number of experimental observations and found to be 6.63W m^{-2} k^{-1}.*

Modeling using python was done in order to understand the temperature and heat loss profiles over time using the parameters specified in *Table 3*. The time for the wort temperature to decrease from 72℃ to 62℃ is predicted to be 24m 30s, the thermal decay is demonstrated in *Figure 4* and the python code is in the appendix. It is useful to note that based on the literature range 2-25 W m^{-2} k^{-1} (Incropera et al., 2011), the predicted times are inversely correlated ranging from 38m 12s for 2 W m^{-2} k^{-1} and 10m 7s for 25 W m^{-2} k^{-1} which corresponds to a margin of 73.71%.

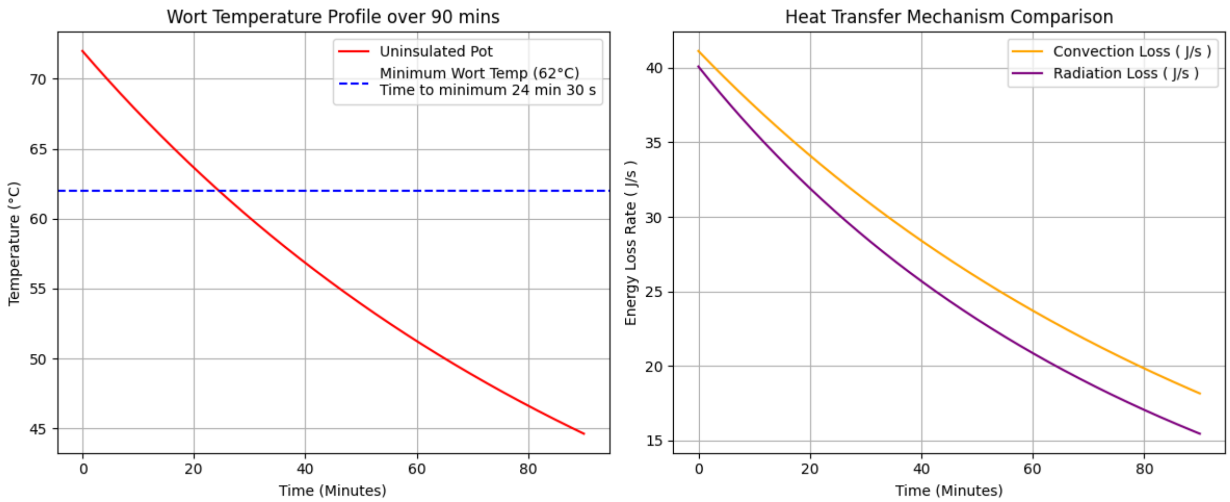

Figure 4 - Predicted thermal decay for uninsulated pot

This effectively allows optimal energy usage by predicting how long the wort remains in the enzyme optimal temperature without inputting any power from the stove which is useful in cost reduction (negligible in small scale production but becomes significant with larger production scales) and preserves induction coils from degradation involved with heating for long times.

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Appendix

Thermal Modeling

```
import matplotlib.pyplot as plt

# Constants
thermal_conductivity = 16.2  # W/(m K)
Cp_ss = 500  # J/(kg K)
Cp_water = 4184  # J/(kg K)
density_ss = 7930  # kg/m3
density_water = 1000  # kg/m3
emissivity_ss304 = 0.8
emissivity_water = 0.963
heat_transfer_coef = 6.63  # W/(m2 k)
T_env_C = 23  # Celcius
T_env_K = T_env_C + 273.15  # Kelvin
stephan_boltzman_coeff = 5.67e-8  # W/(m2 K4)
A_side = 0.0806  # m2
A_top = 0.046  # m2

T_start_C = 72.0
simulation_minutes = 90
dt = 1
total_steps = simulation_minutes * 60

# total heat capacity
mass_ss = (21.324 / 1e6) * density_ss  # kg (pot mass)
mass_water = 0.0025 * density_water  # kg (2.5l water)
```

```
total_heat_capacity = (mass_ss * Cp_ss) + (mass_water * Cp_water)

#plotting lists
time_axis = []
temperature_history = []
fc_loss_history = []
rad_loss_history = []

current_T_C = T_start_C

for step in range(total_steps):
   current_time_seconds = step * dt
   current_T_K = current_T_C + 273.15
   # free convection
   dT = current_T_C - T_env_C
   q_fc = heat_transfer_coef * (dT * A_side + dT * A_top)  # j/s

   # radiation
   q_rad = stephan_boltzman_coeff * ((current_T_K**4 - T_env_K**4) *
emissivity_ss304 * A_side + (current_T_K**4 - T_env_K**4) * emissivity_water *
A_top)  # j/s

   # Total loss
   total_power_lost = q_fc + q_rad

   # Instantaneous temperature drop
   # Q = P * t   dT = Q / Cp_total
   energy_lost = total_power_lost * dt
   temperature_drop = energy_lost / total_heat_capacity

   current_T_C -= temperature_drop

   time_axis.append(current_time_seconds / 60.0)
   temperature_history.append(current_T_C)
   fc_loss_history.append(q_fc)
   rad_loss_history.append(q_rad)
   if 61.99 < temperature_history[step] < 62 :
       SEC = (step/60 - step//60)*60
       MIN = step/60 - (step/60 - step//60)

plt.figure(figsize=(12, 5))

# Temperature Decay Curve
plt.subplot(1, 2, 1)
plt.plot(time_axis, temperature_history, color="red", label="Uninsulated Pot")
plt.axhline(62, color="blue", linestyle="--", label=f"Minimum Mash Temp
(62°C)\nTime to minimum {int(MIN)} min {int(SEC)} s")
plt.title("Mash Temperature Profile over 1 Hr")
plt.xlabel("Time (Minutes)")
```

```
plt.ylabel("Temperature (°C)")
plt.grid(True)
plt.legend()

# Convection vs Radiation graph
plt.subplot(1, 2, 2)
plt.plot(time_axis, fc_loss_history, label="Convection Loss ( J/s )",
color="orange")
plt.plot(time_axis, rad_loss_history, label="Radiation Loss ( J/s )",
color="purple")
plt.title("Heat Transfer Mechanism Comparison")
plt.xlabel("Time (Minutes)")
plt.ylabel("Energy Loss Rate ( J/s )")
plt.grid(True)
plt.legend()

plt.tight_layout()
plt.show()
```

Kinetic Modeling

```
import matplotlib.pyplot as plt
import numpy as np
from dataclasses import dataclass

@dataclass
class Batch:
   name: str
   OG: float
   FG: float
   temperature: float # C Max Fermentation
   V: float # L Batch Volume
   mu_max: float # 1/h Max growth rate Voss Kveik
   Ks: float # g/L Half sat const
   Ki: float # L/g Ethanol Inhibition Const
   Yxs: float # g/g Biomass Yield Coeff
   pitching_rate: float # g/L
   lag: float # hrs apparent lag time
   Rmax: float # g/(L h) Max rate of sugar consumption

batch1 = Batch("Batch 1", 1.044, 1.010, 27, 1.7185, 0.45, 0.025, 0.015, 0.1,
1.164, 1.5, 2.8494)
batch2 = Batch("Batch 2", 1.042, 1.008, 27, 2.729, 0.45, 0.025, 0.015, 0.1,
0.73, 2.2, 1.1417)

def data_processing(batch):
   OG = batch.OG
   FG = batch.FG
   C0 = 2714 * OG - 2713
   C1 = 2714 * FG - 2713
   P = C0 - C1
   ABV = ( OG - FG ) * 131.25
   E1 = 7.89 * ABV
   Yes = E1 / P
   return C0, P, E1, Yes
def Gompertz_model(time, batch):
   E_max = E1
   S_t = S_max * np.exp(-1 * np.exp(((batch.Rmax * np.e) / (S_max)) *
(batch.lag - time) + 1))
   E_t = E_max * np.exp(-1 * np.exp(((batch.Rmax * np.e) / (E_max)) *
(batch.lag - time) + 1))
   return S_t, E_t
def Monod_Growth_model(time, batch):
   C = C0
   X = batch.pitching_rate
   E = 0

   C_history = []
```

```
    X_history = []
    E_history = []

    dt = time[1] - time[0]

    for t in time:
        I = (1 - E / 94.71) ** 2
        mu = batch.mu_max * (C/(batch.Ks + C)) * I # Yeast growth rate
        dX = mu * X * (1 - X/8) * dt
        R = batch.Rmax * C / (batch.Ks + C) * I
        dC = - R * X * dt
        dE = - Yes * dC

        C += dC
        E += dE
        X += dX
        C = max(C, 0)
        C_history.append(C0-C)
        E_history.append(E)
        X_history.append(X)
    return C_history, X_history, E_history
def main(batch, timescale=24):
    time = np.linspace(0,timescale, timescale + 1)
    S_t = []
    E_t = []
    for i in time:
        y1, y2 = Gompertz_model(i, batch)
        S_t.append(y1)
        E_t.append(y2)

    sugar, biomass, E = Monod_Growth_model(time, batch)

    plt.figure(figsize=(10,7))

    plt.subplot(2, 1, 1)
    plt.plot(time,S_t, label=f"{batch.name} \n Sugar Consumed {round(((max(S_t)
- min(S_t)) * batch.V ), 1)} g")
    plt.plot(time, E_t, label=f"{batch.name} Ethanol Produced")
    plt.xlim(min(time))
    plt.ylim(min(S_t))
    plt.title("Modified Gompertz Model")
    plt.xlabel("Time (hrs)")
    plt.ylabel("Concentration (g/L)")
    plt.grid(True)
    plt.legend()

    plt.subplot(2, 1, 2)
    plt.plot(time, sugar, label=f"{batch.name} \n Sugar Consumed
{round(((max(sugar) - min(sugar)) * batch.V ), 1)} g")
```

```
    plt.plot(time, E, label=f"{batch.name} Ethanol Produced")
    plt.xlim(min(time))
    plt.ylim(min(E))
    plt.title("Modified Monod-Logistic Model")
    plt.xlabel("Time (hrs)")
    plt.ylabel("Concentration (g/L)")
    plt.grid(True)
    plt.legend()

    plt.subplots_adjust(wspace=0.1, hspace=0.5)
    plt.show()

command = int(input(f"Which batch are you looking for? \nType in the corresponding batch number - "))
corresponding = {1: batch1, 2: batch2}
requested_timescale = int(input(f"What modelling timescale are you looking for? \n Type a number in hours - "))

C0, S_max, E1, Yes = data_processing(corresponding[command])
main(corresponding[command], requested_timescale)
```